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Assembled battery

Abstract

A technology is provided which suppresses the deterioration of the joint part between the electrode terminal and the bus bar due to an external load. An assembled battery herein disclosed includes a plurality of single cells, and a bus bar. The plurality of single cells each include an electrode body, a battery case, and an electrode terminal. The electrode terminal in at least one single cell among the plurality of single cells includes a first member having one end arranged outside the battery case, and the other end arranged inside the battery case, and a second member arranged at the outer surface of the battery case, and mounted at the end of the first member. Herein, a flexible part protruding toward the bus bar is formed at the surface of the second member, and the electrode terminal is joined with the bus bar at the flexible part.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the priority based on Japanese Patent Application No. 2021-047151 filed on Mar. 22, 2021, the entire contents of which are incorporated by reference in the present specification.

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(2) The present invention relates to an assembled battery.

2. Background

(3) Currently, a secondary battery such as a lithium ion secondary battery or a nickel hydrogen battery has been widely used in various fields including a vehicle, a portable terminal, and the like.

Such a secondary battery generally includes an electrode body which is a power generating element and a battery case for accommodating the electrode body. Further, for such a secondary battery, such a structure is adopted, which leads out the electrode terminal connected with the electrode body inside the battery case to outside the case. Then, the secondary battery of the foregoing structure can be mounted on a vehicle or the like while configuring an assembled battery. Such an assembled battery is constructed by arraying a plurality of secondary batteries along a prescribed array direction, and connecting respective electrode terminals of the plurality of arrayed secondary batteries by a bus bar. In the following description, the secondary battery forming an assembled battery is also referred to as a “single cell”.

(4) Examples of the related art technology regarding the junction between the electrode terminal and the bus bar may include WO 2017/130705. In this publication, in bus bars connecting (joining) a battery cell and a battery cell arranged adjacent to each other, a connection is established between a first connection part and a second connection part connected to one battery cell and another battery cell, respectively via folded parts at 3 sites. Appropriate deformation of the folded parts can absorb the deviation between the battery cells, or the like, which can keep the electrical connection state.

SUMMARY OF THE INVENTION

(5) Incidentally, the assembled battery with the foregoing configuration may be applied with an external load such as an impact or vibration on respective constituent members (e.g., the single cells) according to the usage environment or the manufacturing environment. When such an external load acts on the joint part between the electrode terminal and the bus bar, the joint part may be deteriorated, which may result in the reduction of the junction strength, the electric conductivity, or the like.

(6) The present invention has been made in order to solve such a problem. It is an object of the present invention to provide a technology of suppressing the deterioration of the joint part between the electrode terminal and the bus bar due to an external load.

(7) An assembled battery herein disclosed includes a plurality of single cells arrayed along a prescribed array direction, and a bus bar establishing an electrical connection between the plurality of arrayed single cells. The plurality of single cells each include an electrode body, a battery case for accommodating the electrode body, and an electrode terminal electrically connected to the electrode body, and joined with the bus bar. The electrode terminal in at least one single cell among the plurality of single cells includes: a first member made of a metal having one end arranged outside the battery case, and the other end arranged inside the battery case, and a second member made of a metal arranged at an outer surface of the battery case, and mounted at the one end of the first member. Herein, a flexible part protruding toward the bus bar is formed at a surface of the second member, and the electrode terminal is joined with the bus bar at the flexible part.

(8) For the assembled battery with such a configuration, the bus bar is joined with the flexible part protruding from the surface of the second member toward the bus bar. As a result of this, when the constituent member (such as a single cell) of the assembled battery is applied with an external load, the flexible part is deformed so as to bend. For this reason, it is possible to suppress the application of a stress to the joint part between the second member and the bus bar due to an external load. As a result, it is possible to suppress the deterioration of the joint part between the electrode terminal and the bus bar due to an external load.

(9) In accordance with a preferable aspect of the assembled battery herein disclosed, in a cross sectional view including the joint part forming junction between the flexible part and the bus bar, the flexible part has a flat part extending along a surface of the second member, and a pair of wall parts rising from the surface of the second member toward the flat part. As a result of this, the stress due to an external load can be properly absorbed at the flexible part. For this reason, it is possible to more properly suppress the deterioration of the joint part between the electrode terminal and the bus bar.

(10) In accordance with another preferable aspect of the assembled battery herein disclosed, a thickness of the pair of wall parts is smaller than a thickness in a region except for the flexible part of the second member. With such a configuration, the flexible part becomes more likely to bend with respect to an external load. For this reason, it is possible to better suppress the deterioration of the joint part between the electrode terminal and the bus bar.

(11) Further, in accordance with a still other preferable aspect of the assembled battery herein disclosed, a region surrounded by the flat part and the wall parts is a space. By thus providing the space in the inside of the flexible part, the flexible part becomes more likely to bend with respect to an external load. For this reason, it is possible to better suppress the deterioration of the joint part between the electrode terminal and the bus bar.

(12) Still further, in an aspect in which the space is provided in the inside of the flexible part, it is preferable that a contact interface between the first member and the second member is subjected to metal joining. In the aspect in which the space is provided in the inside of the flexible part, in addition to the effect of suppressing the deterioration of the joint part, the present invention can also contribute to the suppression of the thermal deterioration of the metal joint part between the first member and the second member.

(13) Furthermore, in the aspect in which the first member and the second member are joined with each other, it is preferable that the first member and the second member are composed of mutually different metals. When the first member and the second member include different kinds of metals, the deterioration of the metal joint part between the two members tends to be more likely to be caused than when the members include the same kind of metal. However, by providing the space in the inside of the flexible part as in the foregoing aspect, it is possible to suppress the thermal deterioration of the metal joint part between the first member and the second member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view schematically showing an assembled battery in accordance with First Embodiment;
- (2) FIG. 2 is a perspective view schematically showing a single cell of the assembled battery in accordance with First Embodiment;
- (3) FIG. 3 is a plan view for schematically illustrating a joint part between an electrode terminal and a bus bar in the assembled battery in accordance with First Embodiment;
- (4) FIG. 4 is a local cross sectional view for schematically showing a joint part between an electrode terminal and a bus bar in the assembled battery in accordance with First Embodiment;
- (5) FIG. 5 is a cross sectional view for schematically showing a configuration of an electrode terminal for use in the assembled battery in accordance with First Embodiment;
- (6) FIG. 6 is a plan view for schematically showing a joint part between an electrode terminal and a bus bar in an assembled battery in accordance with Second Embodiment;
- (7) FIG. 7 is a cross sectional view for schematically showing a configuration of an electrode terminal for use in an assembled battery in accordance with Third Embodiment; and
- (8) FIG. 8 is a cross sectional view for schematically showing a configuration of an electrode terminal for use in an assembled battery in accordance with Fourth Embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) Below, embodiments of the technology herein disclosed will be described. Incidentally, the following embodiments are not intended to limit the technology herein disclosed. Further, matters necessary for executing the technology herein disclosed, except for matters specifically referred to in the present specification can be grasped as design matters of those skilled in the art based on the related art in the present field. Namely, the technology herein disclosed can be executed based on

the contents disclosed in the present specification, and the technical common sense in the present field.

(10) Incidentally, in the drawings referred to in the following description, the members/parts producing the same action are given the same numeral and sign. Further, the dimensional relationships (such as the length, width, and thickness) in each drawing do not reflect the actual dimensional relationships. Then, in each drawing, a reference sign X represents “width direction”, a reference sign Y represents “depth direction”, and a reference sign Z represents “height direction”. However, the directions are defined for convenience of description, and are not intended to limit the setting form of a secondary battery or an assembled battery in use or during manufacturing. Further, in the present specification, the expression of “A to B” indicating the numerical value range includes the meaning of “A or more and B or less”, as well as the meaning of “more than A and less than B”.

(11) Further, the term “secondary battery” in the present specification denotes an electric storage device in which a charging/discharging reaction is effected by transfer of electric charge carriers between a pair of electrodes (a positive electrode and a negative electrode) via an electrolyte in general. Such a secondary battery also includes a capacitor such as an electric double layer capacitor other than a so-called storage battery such as a lithium ion secondary battery, a nickel hydrogen battery, or a nickel cadmium battery. The technology herein disclosed is not limited to an assembled battery constructed using a specific kind of secondary battery, and is applicable to an assembled battery constructed using a secondary battery having an electrode terminal without particular restriction.

First Embodiment

(12) Below, an assembled battery in accordance with one embodiment (which will be hereinafter referred to as “First Embodiment”) of the technology herein disclosed will be described with reference to FIGS. 1 to 5. FIG. 1 is a perspective view schematically showing an assembled battery in accordance with First Embodiment. FIG. 2 is a perspective view schematically showing a single cell of the assembled battery in accordance with First Embodiment. FIG. 3 is a plan view for schematically illustrating a joint part between an electrode terminal and a bus bar in the assembled battery in accordance with First Embodiment. FIG. 4 is a local cross sectional view for schematically illustrating a joint part between an electrode terminal and a bus bar in the assembled battery in accordance with First Embodiment. FIG. 5 is a cross sectional view for schematically illustrating a configuration of the electrode terminal for use in the assembled battery in accordance with First Embodiment.

(13) Assembled Battery

(14) As shown in FIG. 1, an assembled battery **100** in accordance with the present embodiment includes a plurality of single cells **1** arrayed along a prescribed array direction. Specifically, for the assembled battery **100**, the plurality of single cells **1** each having a flat square battery case **10** (see FIG. 2) are arrayed along the depth direction Y so that respective flat surfaces are opposed to each other. Then, a pair of end plates **120** are arranged on the opposite outer sides in the array direction (depth direction Y) of such an assembled battery **100**. The pair of end plates **120** are crosslinked by a binding beam material **130**, thereby binding each of the plurality of single cells **1** along the array direction. Incidentally, in the present embodiment, in order to equalize the binding pressure in the flat surface, a spacer **140** is arranged between the adjacent single cells **1**. Then, for the assembled battery **100** in accordance with the present embodiment, an electrical connection between the plurality of arrayed single cells **1** is established by bus bars **110**.

(15) Bus Bar

(16) The bus bar **110** is a plate-shaped conductive member as shown in FIG. 1. Although particularly described later, respective single cells **1** are arrayed so that the electrode terminals **20** are adjacent to each other in the depth direction Y, and the bus bar **110** is arranged so as to bridge over the adjacent electrode terminals **20**. Then, respective single cells **1** are electrically connected

with each other via the bus bars **110**, thereby constructing the assembled battery **100**. As shown in FIG. **4**, the bus bar **110** and the electrode terminal **20** are joined in surface contact with each other. Incidentally, although not particularly restricted, as the constituent material for the bus bar **110**, a metal material excellent in conductivity and strength is used. From the viewpoint of suppressing the reduction of the conductivity, for the bus bar **110**, the same kind of metal material as that of a second member **40** of the electrode terminal **20** described later is preferably used. As preferable constituent materials for the bus bar **110**, mention may be made of aluminum, an alloy mainly including aluminum, and the like. Further, the shape of the bus bar **110** has no particular restriction, and a rod-shaped bus bar may be used in place of a sheet-shaped bus bar.

(17) Single Cell

(18) Then, a description will be given to a configuration of the single cell **1** for use in the assembled battery **100** having the foregoing configuration. Incidentally, the number of the single cells **1** constructing the assembled battery **100** has no particular restriction, and can be appropriately changed according to the object of the assembled battery **100** (the required electric power or the standard of an external device, or the like). Further, with the technology herein disclosed, all of the plurality of single cells constructing the assembled battery are not required to have the following configuration. Namely, it is essential only that the assembled battery herein disclosed includes the single cell with the following configuration as at least one of the single cells constructing the assembled battery. In other words, the assembled battery herein disclosed may include a single cell having a different configuration from that of the single cell configured as described below.

(19) Overall Configuration

(20) As shown in FIG. **2**, the single cell **1** includes an electrode body (not shown), a battery case **10**, and an electrode terminal **20**.

(21) Electrode Body

(22) The electrode body is a power generating element accommodated in the inside of the battery case **10**. The structure of the electrode body has no particular restriction, and various structures adoptable in a general secondary battery can be adopted without particular restriction. For example, for the electrode body, the structure in which the positive electrode and the negative electrode are stacked one on another via the separator can be adopted. Specific examples of the structure of this kind of electrode body may include a wound electrode body in which long band-shaped positive electrode, negative electrode, and separator are stacked one on another in a wound manner, and a lamination type electrode body in which rectangular sheet-shaped positive electrode, negative electrode, and separator are stacked one on another. Incidentally, for the particular structures and materials of respective members (the positive electrode, the negative electrode, and the separator) forming the electrode body, those adoptable in a general secondary battery (e.g., a lithium ion secondary battery) can be adopted without particular restriction, and do not limit the technology herein disclosed. For this reason, a detailed description thereon will be omitted.

(23) Electrolyte

(24) Further, although not shown, for the single cell **1**, other than the electrode body, the electrolyte is also accommodated in the inside of the battery case **10**. As the electrolytes, a nonaqueous type liquid electrolyte (nonaqueous electrolyte) including a nonaqueous solvent and a support salt, a solid electrolyte obtained by forming a powder-shaped electrolyte into a sheet shape, and the like can be used. Incidentally, the specific components of the electrolyte do not limit the technology herein disclosed, and hence will not be described in details.

(25) Battery Case

(26) The battery case **10** is a container for accommodating the electrode body and the electrolyte therein. As shown in FIG. **2**, the battery case **10** is a flat square container. Such a square battery case **10** includes a flat square case main body **12** opened at the upper surface thereof, and a sheet-shaped lid body **14** for blocking the upper surface opening of the case main body **12**. Further,

although described in details later, a terminal insertion hole of the opening for inserting each of a pair of electrode terminals **20** therethrough is provided at each opposite end of the lid body **14** in the width direction X. Incidentally, the outer shape of the battery case **10** is not limited to the foregoing shape, and can be appropriately changed according to the specifications of an external device, the shape of the electrode body, or the like. For example, the battery case **10** may be a bottomed cylindrical case. Further, the material for the battery case **10** has no particular restriction so long as it is a material having a prescribed strength. Preferable examples of the material for the battery case **10** may include a metal material which is lightweight, and has good thermal conductivity (e.g., aluminum, stainless steel, or nickel plating steel).

(27) Electrode Terminal

(28) As shown in FIG. **1**, each of the plurality of single cells **1** has a pair of electrode terminals **20** to be joined with the bus bar **110**. As shown in FIG. **4**, each electrode terminal **20** has a first member **30** and a second member **40**. Although not shown, the electrode terminal **20** is electrically connected with the electrode body in the inside of the battery case **10**. Of the pair of electrode terminals **20**, in the inside of the battery case **10**, the electrode terminal electrically connected with the positive electrode is referred to as a positive electrode terminal, and the electrode terminal electrically connected with the negative electrode is referred to as a negative electrode terminal. The structure of the electrode terminal **20** described below may be adopted for any one of or both of the positive electrode terminal and the negative electrode terminal.

(29) First Member

(30) The first member **30** is a member made of a metal which is arranged at one end thereof in the outside of the battery case **10**, and is arranged at the other end thereof in the inside of the battery case **10**. Specifically, the first member **30** includes a flat sheet-shaped base part **34**, and a pillar-shaped shaft part **32**. The base part **34** forms one end of the first member **30**, is arranged in the outside of the lid body **14** of the battery case **10**, and is arranged along the outer surface of the lid body **14**. The planar shape of the base part **34** has no particular restriction, and may be a circular shape (which includes a generally circular shape and elliptic shape; the same shall apply hereinafter.), or may be a rectangular shape (which includes a generally rectangular shape: the same shall apply hereinafter). The size of the base part **34** in a plan view may only be a size enough to block an insertion hole **14a** of the lid body **14**. The shaft part **32** extends from the base part **34** in the vertical direction Z, and penetrates through the lid body **14**. The outer shape of the shaft part **32** has no particular restriction, and may be a cylindrical shape, or may be a prismatic shape. Then, the lower end of the shaft part **32** forms the other end of the first member **30**, and is arranged in the inside of the battery case **10**. At the lower end of the shaft part **32**, a rivet part **36** is formed. The rivet part **36** is formed by caulking processing for pressing and deforming the lower end of the shaft part **32** formed in a tubular shape toward the radial outside. The rivet part **36** fixes a third member **50** to the battery case **10** (the lid body **14**), and electrically connect the first member **30** and the third member **50**.

(31) Although not particularly restricted, the first member **30** can include aluminum, an alloy mainly including aluminum, copper, or an alloy mainly including copper. Incidentally, in the present specification, the term “alloy mainly including aluminum” represents an alloy including aluminum in an amount of at least 70% or more. On the other hand, the term “alloy mainly including copper” represents an alloy including copper in an amount of at least 50% or more. Incidentally, other elements which can be included in the first member **30** have no particular restriction, and mention may be made of silicon, iron, manganese, magnesium, zinc, chromium, titanium, lead, zirconium, or the like.

(32) Second Member

(33) The second member **40** is a member made of a metal arranged on the outer surface of the battery case **10**, and mounted at one end of the first member **30**. As shown in FIGS. **2**, **4**, and **5**, the second member **40** is a sheet-shaped member arranged along the surface of the lid body **14** in the

outside of the battery case **10** (lid body **14**). A concave part **42** is formed in a first surface **40a** (the surface on the battery case **10** side) of the second member **40**, and the base part **34** of the first member **30** is fitted into the concave part **42**. In the inside of the concave part **42**, the first member **30** (base part **34**) and the second member **40** are in surface contact with each other. Although not particularly restricted, the contact interface between the first member **30** and the second member **40** is preferably subjected to metal joining. As one example, the contact interface between the first member **30** and the second member **40** situated on a shaft center SC of the shaft part **32** may be desirably joined, thereby forming a metal joint part **70**. As a result of this, when the electrode terminal **20** is applied with an external load, the second member **40** can be regulated from moving in the width direction X or the depth direction Y. The means for the joining has no particular restriction, and conventionally known joining means such as ultrasonic welding, laser welding, or resistance welding may be appropriately used. Incidentally, although described in details later, as in the present embodiment, by forming a flexible part **44** having a space **44c** in the inside thereof, it is also possible to contribute to the suppression of the thermal deterioration of the metal joint part **70**. Further, although not particularly restricted, if required, a lock part **46** protruding toward the shaft part **32** of the first member **30** may be provided at the lower end of the sidewall of the concave part **42**. This can regulate the movement of the second member **40** in the height direction Z. Incidentally, the planar shape of the second member **40** has no particular restriction, and may be a circular shape, or may be a rectangular shape.

(34) Further, the metal forming the second member **40** may be the same as, or may be different from the metal forming the first member **30**. When the first member **30** and the second member **40** include mutually different metals, respectively, the metal joint part **70** is formed by dissimilar metal joining. Such dissimilar metal joining tends to cause deterioration due to heat. For this reason, the effect resulting from the formation of the flexible part **44** having the space **44c** in the inside thereof is further preferably exhibited. Incidentally, when the first member **30** and the second member **40** include dissimilar metals, mention may be made of the configuration in which the first member **30** includes a copper type material, and the second member **40** includes an aluminum type material.

(35) The flexible part **44** is formed at a second surface **40b** (the surface opposite to the first surface **40a**) of the second member **40**. The flexible part **44** protrudes toward the bus bar **110** as shown in FIG. 4. Then, the electrode terminal **20** in the present embodiment is joined with the bus bar **110** at the flexible part **44** of the second member **40**. Such joining means is typically laser welding.

Namely, the joint part **90** between the second member **40** and the bus bar **110** is typically a thermally welded part. The joint part **90** is formed so as to penetrate through the bus bar **110** in the height direction Z, and reach the flexible part **44**. For the assembled battery **100** in accordance with the present embodiment, when the constituent member (such as the single cell **1**) of the assembled battery **100** is applied with an external load, the flexible part **44** is deformed in a bending manner. For this reason, it is possible to suppress the application of a stress due to an external load on the joint part **90** between the second member **40** and the bus bar **110**. As a result of this, it is possible to suppress the deterioration of the joint part **90** between the electrode terminal **20** and the bus bar **110** due to an external load.

(36) Below, a description will be given to one example of the specific structure of the flexible part **44**. As shown in FIG. 3, the flexible part **44** is a convex-shaped member formed in a ring shape with the shaft part **32** of the first member **30** (the shaft center SC in FIG. 4) as the center in a plan view. The region surrounded by the bus bar **110**, the second surface **40b** of the second member **40**, and the flexible part **44** becomes the space (reference sign S). By forming the flexible part **44** in such a planar ring shape, it is possible to sufficiently ensure the contact area with the bus bar **110**, and to improve the junction strength. Further, as shown in FIG. 4, in a cross sectional view including the joint part **90** between the flexible part **44** and the bus bar **110**, the flexible part **44** has a flat part **44a** extending along the surface (second surface **40b**) of the second member **40**, and a pair of wall parts **44b** rising from the surface (second surface **40b**) of the second member **40** toward

the flat part **44a**. The flexible part **44** is joined at the flat part **44a** with the bus bar **110** in a surface contact state. For the flexible part **44** with such a configuration, when a stress due to an external load applies thereto, the pair of wall parts **44b** are deformed, which can absorb the stress. For this reason, it is possible to more properly suppress the deterioration of the joint part **90** due to application of a strong stress thereto.

(37) Further, for the assembled battery **100** in accordance with the present embodiment, the flexible part **44** for absorbing the external load is formed not at the bus bar **110**, but at the second member **40** of the electrode terminal **20**. As a result of this, it is possible to suppress the increase in height dimension of the overall assembled battery **100**. Further, when concave and convex are formed on the surface of a bus bar, joining (such as laser welding) between the bus bar and an electrode terminal may be hindered. In contrast, in the present embodiment, provision of the flexible part **44** on the side of the electrode terminal **20** can flatten the surface of the bus bar **110**. This can facilitate joining between the bus bar **110** and the electrode terminal **20**.

(38) Incidentally, for the second member **40** in the present embodiment, a space **44c** is formed in the region surrounded by the flat part **44a** and the wall parts **44b**. More specifically, the space **44c** is the region surrounded by the flat part **44a**, the wall parts **44b**, and the second surface **40b**. The space **44c** is thus formed in the inside of the flexible part **44**, thereby making the flexible part **44** more likely to bend with respect to an external load. For this reason, it is possible to more preferably prevent the deterioration of the joint part **90**. Further, the space **44c** in the inside of the flexible part **44** can also function as a heat insulating space. For this reason, it is possible to prevent the thermal deterioration due to transmission of a large heat to the metal joint part **70** between the first member **30** and the second member **40** when the bus bar **110** and the electrode terminal **20** are subjected to laser welding. Incidentally, from the viewpoint of ensuring the strength of the flexible part **44** to a certain extent, the space **44c** may be filled with a resin material. Such a resin material is preferably a resin material which tends to elastically deform, has excellent flexibility, and is low in rigidity. Examples of the resin material to be filled in the space **44c** may include acrylic resin and elastic rubber.

(39) Below, a description will be given to respective dimensions on the flexible part **44** in the present embodiment. First, as shown in FIG. 5, the width **L2** of the flexible part **44** is preferably set smaller than the width **L1** of the second member **40**. From the viewpoint of properly absorbing the stress due to an external load, and preferably preventing the deterioration of the joint part **90**, the ratio (**L2/L1**) of the width **L2** of the flexible part **44** to the width **L1** of the second member **40** is preferably about $1/10$ to $2/5$. Further, from the viewpoint of suppressing the deterioration of the flexible part **44** due to an external load, the **L2/L1** is more preferably set at, for example, about $1/8$ to $1/4$. Incidentally, as shown in FIG. 5, the term “width **L2** of the flexible part **44**” in the present specification represents the dimension between the end **44k** and the end **44m** of the wall part **44b**.

(40) Further, although not particularly restricted, the width **L3** of the flat part **44a** in the width direction **X** is preferably the same as the width **L2**, or smaller than that. From the viewpoint of more preferably preventing the deterioration of the joint part **90** or the flexible part **44**, the ratio (**L3/L2**) of the width **L3** of the flat part **44a** to the width **L2** of the flexible part **44** is preferably about $1/2$ to $4/5$. Incidentally, the term “width **L3** of the flat part **44a**” in the present specification represents the dimension between the end **44q** and the end **44n** of the flat part **44a**.

(41) Further, the height **H1** of the flexible part **44** is preferably smaller than the thickness **D1** of the second member **40**. As a result of this, it is possible to suppress the deterioration of the flexible part **44** due to an external load. On the other hand, with an increase in height **H1** of the flexible part **44**, the stress applied to the joint part **90** between the electrode terminal **20** and the bus bar **110** tends to decrease. From such a viewpoint, the height **H1** of the flexible part **44** is desirably set at about $1/5$ to $2/3$ of the thickness **D1** of the second member **40**, and preferably set at about, for example, about $1/4$ to $1/2$. Incidentally, the height **H1** of the flexible part **44** is the dimension between the end **44m** and the end **44n**. Further, the thickness **D1** of the second member **40** is the dimension from the first

surface **40a** to the second surface **40b**, and is the thickness except for the height of the flexible part **44**.

(42) Further, the inclination angle θ of the wall parts **44b** can be appropriately set between 70° to 90° . For example, as the inclination angle θ of the wall parts **44b** approaches 90° , the wall parts **44b** becomes more likely to absorb the external load in the horizontal direction (the width direction X and the depth direction Y). On the other hand, with a decrease in inclination angle θ of the wall parts **44b**, it becomes easier to absorb the external load in the height direction Z at the wall parts **44b**. Incidentally, the inclination angle θ of the wall parts **44b** in the the present specification represents the more acute angle of the angles formed between the second surface **40b** of the second member **40** and the wall parts **44b**.

(43) The curvature radius R of the portion of the wall part **44b** rising from the second surface **40b** of the second member **40** has no particular restriction, and may be 0.05 mm to 1 mm. Incidentally, from the viewpoint of properly absorbing the external load, the curvature radius R of the rising part is preferably, for example, 0.1 mm to 0.8 mm. Incidentally, the “curvature radius R of the rising part” is the radius of a virtual circle in contact with the second surface **40b** of the second member **40** and the wall part **44b**.

(44) Further, the thickness D3 of the wall part **44b** is preferably set comparable to the thickness D1 of the second member **40**, or smaller than the thickness D1. As a result of this, the flexible part **44** becomes more likely to bend. For this reason, it is possible to more preferably prevent the deterioration of the joint part **90**. As one example, the thickness D3 is desirably set at about $\frac{1}{5}$ to $\frac{2}{3}$ of the thickness D1, and is preferably set at, for example, about $\frac{1}{4}$ to $\frac{1}{2}$. On the other hand, the thickness D2 of the flat part **44a** has no particular restriction. However, the thickness D2 of the flat part **44a** is preferably 0.5 mm or more. As a result of this, when the bus bar **110** and the flexible part **44** are joined, the joint part **90** can be prevented from penetrating through the flat part **44a**.

(45) Third Member

(46) Incidentally, the constituent members of the electrode terminal **20** are not limited to the first member **30** and the second member **40**. For example, the electrode terminal **20** in the present embodiment includes the third member **50** as shown in FIG. 4. Although not shown, the third member **50** has a collector part extending along the height direction Z. The collector part of the third member **50** is connected at the lower end thereof with the electrode body (the positive electrode or the negative electrode). Then, as shown in FIG. 4, the upper end **52** of the third member **50** is bent so as to be along the inner surface of the battery case **10** (lid body **14**). Then, the upper end **52** of the third member **50** is fixed to the battery case **10** (lid body **14**) by the rivet part **36** of the first member **30**, and is electrically connected with the first member **30**. As a result of this, the electrode body in the battery case **10** and the second member **40** outside the battery case **10** are electrically connected with each other via the third member **50** and the first member **30**.

Incidentally, the third member **50** is not an essential constituent member of the electrode terminal **20**. For example, when a part of the first member **30** is extended along the height direction, and is connected with the electrode body, an electrode terminal **20** not having the third member **50** can be constructed.

(47) Insulating Material

(48) Incidentally, the single cell of the assembled battery herein disclosed may include an insulating material for preventing the conduction between the electrode terminal **20** and the battery case **10**. As one example, the single cell **1** includes a gasket **62** and insulator **64** as insulating materials.

(49) Gasket

(50) The gasket **62** is an insulating material arranged on the outer surface of the battery case **10** (the upper surface of the lid body **14**). Such a gasket **62** is a box-shaped insulating material having an accommodating part **62a** on the upper surface side thereof. The electrode terminal **20** (the base part **34** of the first member **30** and the second member **40**) is accommodated in the accommodating part

62a of the gasket **62**. As a result of this, an insulation is established between the lid body **14** and the first member **30**, and between the lid body **14** and the second member **40**. Further, a tubular-shaped convex part **62b** having a terminal insertion hole **62c** is formed at the gasket **62** in the present embodiment. The convex part **62b** of the gasket **62** is inserted into the terminal insertion hole **14a** of the lid body **14**. Then, the convex part **62b** of the gasket **62** is interposed between the shaft part **32** of the first member **30** in the inside of the terminal insertion hole **14a** and the lid body **14**, which can insulate the members. Further, the gasket **62** is pressurized at the caulking processing for forming the rivet part **36** of the first member **30**, and hence is compressed between the lid body **14** and the second member **40** (or between the lid body **14** and the base part **34** of the first member **30**). As a result of this, the gap upon mounting each member on the lid body **14** is sealed, which can prevent circulation of liquid (penetration of moisture, and leakage of the electrolyte) inside and outside the battery case **10**. Incidentally, the gasket **62** is preferably formed of an insulating resin which tends to be elastically deformed. Examples of the material for such a gasket **62** may include a fluorinated resin such as perfluoroalkoxy fluorine resin (PFA), a polyphenylene sulfide resin (PPS), and aliphatic polyamide.

(51) Insulator

(52) The insulator **64** is a sheet-shaped insulating material having an opening **64a**, and is arranged between the battery case **10** (lid body **14**) and the third member **50**. As a result of this, the lid body **14** and the third member **50** are insulated from each other. Then, the opening **64a** of the insulator **64** is arranged so as to overlap the terminal insertion hole **62c** of the gasket **62** in a plan view, so that the shaft part **32** of the first member **30** is inserted therethrough. As a result of this, it is possible to prevent the contact between the shaft part **32** of the first member **30** and the lid body **14** with reliability. Further, at the caulking processing for forming the rivet part **36** at the lower end of the first member **30**, the lower surface of the convex part **62b** of the gasket **62** is pressed against the circumferential part of the opening **64a** of the insulator **64**. As a result of this, the gasket **62** and the insulator **64** are crimped. For this reason, it is possible to prevent the circulation of the liquid inside and outside the battery case **10** with more reliability. Incidentally, the insulator **64** is preferably formed of an insulating resin which tends to be elastically deformed as with the gasket **62**.

OTHER EMBODIMENTS

(53) Up to this point, one embodiment of the technology herein disclosed has been described. Incidentally, the First Embodiment shows one example of the assembled battery to which the technology herein disclosed is applied, and is not intended to limit the technology herein disclosed. Below, other embodiments of the technology herein disclosed will be described. Incidentally, in the following description, generally the same configuration as that of the assembled battery **100** in accordance with the First Embodiment can be adopted except for the points specifically referred to.

Second Embodiment

(54) For example, as shown in FIG. 3, in First Embodiment, the flexible part **44** in a planar ring shape was formed at the second member **40**. However, the planar shape of the flexible part is not limited thereto. FIG. 6 is a plan view for schematically illustrating the joint part between the electrode terminal and the bus bar in an assembled battery in accordance with Second Embodiment. As shown in FIG. 6, a second member **240** of the electrode terminal **220** in Second Embodiment has a flexible part **244** in a linear shape in a plan view. Specifically, the second member **240** is provided with a pair of linear flexible parts **244** opposed to each other with the shaft part **32** of the first member **30** (see FIG. 4) interposed therebetween. Even in such a planar shape, the flexible part **244** can be properly deformed with respect to an external load. For this reason, it is possible to suppress the application of a strong stress to the joint part **290** between the bus bar and the electrode terminal **220**.

Third Embodiment

(55) Further, the flexible part may only protrude toward the bus bar, and is also not required to have the cross sectional structure as described in the First Embodiment. Specifically, in First

Embodiment, there is provided the flexible part **44** including the space **44c** surrounded by the flat part **44a**, the wall parts **44b**, and the second surface **40b** formed in the inside thereof. The electrode terminal **20** with such a configuration is formed by, for example, welding the flexible part **44** to the second surface **40b** of the sheet-shaped second member **40**. However, the cross sectional structure of the flexible part is not limited thereto. FIG. 7 is a cross sectional view for schematically illustrating the electrode terminal for use in an assembled battery in accordance with Third Embodiment. As shown in the FIG. 7, for an electrode terminal **320** in Third Embodiment, a flexible part **344** is formed by bending a sheet-shaped second member **340**. The flexible part **344** in Third Embodiment has a flat part **344a** extending along the second surface **340b** of the second member **340**, and a pair of wall parts **344b** rising from the second surface **340b**, and the contact interface between the concave part **342** and the first member **30** toward the flat part **344a**. Further, when the flexible part **344** with such a configuration is provided, a space **344c** surrounded by the flat part **344a**, the wall parts **344b**, and the first member **30** is formed. Even when the flexible part **244** with such a cross sectional structure is provided, it is possible to suppress the application of a strong stress to the joint part between the bus bar and the electrode terminal. Further, for the electrode terminal **320** in Third Embodiment, the space **344c** is formed around the metal joint part **370** between the first member **30** and the second member **340**. For this reason, it is possible to further preferably suppress the thermal deterioration of the metal joint part **370**. Incidentally, a reference sign **340a** represents the first surface of the second member, and **346** represents the lock part.

Fourth Embodiment

(56) Further, in the First to Third Embodiments, in any case, in a cross sectional view, a pair of flexible parts **44** are formed so as to interpose the shaft part **32** of the first member **30** therebetween. However, the number of the flexible parts to be formed in a cross sectional view has no particular restriction. FIG. 8 is a cross sectional view for schematically illustrating the configuration of the electrode terminal for use in an assembled battery in accordance with Fourth Embodiment. The second member **440** of the electrode terminal **420** in Fourth Embodiment is provided with one flexible part **444** in a cross sectional view. Even when the flexible part **444** with such a configuration is formed, it is possible to suppress the application of a strong stress to the joint part between the bus bar and the electrode terminal. Incidentally, when the number of the flexible parts **444** to be formed in a cross sectional view is set at 1 as in the present embodiment, the width of the flexible part **444** in the width direction X can be elongated. For this reason, it is possible to form the flexible part **444** which is more likely to bend with respect to an external load. Further, a reference sign **440a** represents the first surface of the second member; **440b**, the second surface of the second member; **442**, the concave part; **444a**, the flat part of the the flexible part; **444b**, the wall part of the flexible part **444**; **444c**, the space; **446**, the lock part; and **470**, the metal joint part between the first member **30** and the second member **40**.

OTHER EMBODIMENTS

(57) Further, in the First to Fourth Embodiments, in any case, the flexible part having a flat part and a pair of wall parts is formed. However, the flexible part has no particular restriction on the structure so long as it can be deformed so as to bend with respect to an external load, which can suppress the stress on the joint part between the bus bar and the electrode terminal. For example, the flexible part may include an elastic member (e.g., elastic rubber or spring) mounted on the surface of the second member, and a flat part mounted at the upper end of the elastic member. When the flexible part with such a configuration is provided, bending of the elastic member with respect to an external load can suppress the application of a large stress on the joint part between the flat part and the bus bar.

(58) Further, the structure of other members than the flexible part can be appropriately changed. For example, in the embodiments, in any case, the flat sheet-shaped bus bar **110** is used. However, concave and convex may be provided on the bus bar unless the junction between the bus bar and

the electrode terminal is hindered. For example, such a concave part as to be equivalent to the shape of the flexible part may be formed on the surface on the electrode terminal side of the bus bar (typically, the surface in contact with the flexible part). As a result of this, it is possible to more firmly join the flexible part and the bus bar.

(59) Up to this point, specific examples of the technology herein disclosed have been described in details. However, these are merely illustrative, and should not be construed as limiting the scope of the appended claims. The invention herein disclosed includes various modifications and changes of the specific examples.

Claims

1. An assembled battery, comprising: a plurality of single cells arrayed along an array direction, and a bus bar establishing an electrical connection between the plurality of single cells, the plurality of single cells each comprising an electrode body, a battery case in a flat square shape and accommodating the electrode body, the battery base having a flat surface, and an electrode terminal electrically connected to the electrode body and joined with the bus bar, the plurality of single cells arrayed in the array direction and respective flat surfaces of the plurality of single cells are opposed to each other in the array direction, the bus bar bridging over adjacent electrode terminals of the plurality of single cells in the array direction, the electrode terminal in at least one single cell among the plurality of single cells comprising: a first member made of metal and having one end arranged on an outer side of the battery case, and another end arranged on an inner side of the battery case, and a second member which is sheet shaped and made of a metal arranged at an outer surface of the battery case, and mounted at the one end of the first member, wherein the second member has a first surface on a side of the battery case, and a second surface opposite to the first surface, the second member comprises a flexible part on the second surface, the flexible part protruding from the second surface toward the bus bar, the electrode terminal is joined with the bus bar at the flexible part, a height of the flexible part is smaller than a thickness of a portion of the second member from the first surface to the second surface, in a cross sectional view including a joint part forming a junction between the flexible part and the bus bar, the flexible part comprises a flat part extending in a direction along the second surface of the second member, and a pair of wall parts protruding from the second surface of the second member toward the flat part, a region surrounded by the flat part and the pair of wall parts is a space, and the flexible part is a convex-shaped part in a ring shape in a plan view.

2. The assembled battery according to claim 1, wherein a thickness of the pair of wall parts is smaller than the thickness of the portion of the second member from the first surface to the second surface.

3. The assembled battery according to claim 1, wherein a contact interface between the first member and the second member is subjected to metal joining.

4. The assembled battery according to claim 3, wherein the first member and the second member are composed of mutually different metals.

5. The assembled battery according to claim 1, wherein the space is surrounded by the flat part, the pair of wall parts and the second surface of the second member.
